

NON-SIMPLY-LACED CHIRAL COPRODUCTS VIA σ -ORBIFOLD FOLDING

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ABSTRACT. Diagram folding compares ADE Yangians with non-simply-laced Yangians through folded/orbit RTT presentations. It is not used here as a general fixed-point theorem $Y_h(\mathfrak{g})^\sigma = Y_h(\mathfrak{g}^\sigma)$; that equality is false in this generality. The chiral lift is subtler. Equivariance of the Jindal–Kaubrys–Latyntsev critical CoHA vertex coproduct [1] gives a coproduct on invariants only after the landing condition

$$\Delta(V^\sigma) \subset V^\sigma \otimes V^\sigma$$

is proved. Diagonal σ -equivariance alone gives $\Delta(V^\sigma) \subset (V \otimes V)^\sigma$, which may contain nontrivial tensor sectors. The descent criterion has four parts: potential invariance, Hecke equivariance, finite-orbifold E_i preservation, and an independent sector-vanishing calculation, followed by the spectral-parameter and root-length normalisations needed for an RTT comparison. The Yangian-level $\mathcal{A}_3$ coproduct equivariance is explicit; the chiral folded coproducts are governed by the landing obstruction.

I. SETUP: FOLDING, ORBIFOLD ROUTE, AND THE FOUR CHECKS

Let $\mathfrak{g}^{\text{simply}}$ be a simply-laced simple Lie algebra (A , D , or E) and $\sigma \in \text{Aut}(\text{Dyn}(\mathfrak{g}^{\text{simply}}))$ a Dynkin automorphism of order $d \in \{2, 3\}$. The Langlands-dual folding identifies the σ -invariant simple Lie algebra

$$\mathfrak{g} = (\mathfrak{g}^{\text{simply}})^\sigma$$

with the non-simply-laced simple Lie algebra of the indicated type:

$$B_N = (D_{N+1})^{\mathbb{Z}/2}, \quad C_N = (A_{2N-1})^{\mathbb{Z}/2}, \quad F_4 = (E_6)^{\mathbb{Z}/2}, \quad G_2 = (D_4)^{\mathbb{Z}/3}.$$

(The Lie type and the chosen RTT representation are separate data. The order-two $\mathcal{A}_3$ example below gives the type- C_2 fold, identified with B_2 only after the exceptional isomorphism $\mathfrak{sp}_4 \simeq \mathfrak{so}_5$.)

The affine level on the folded current algebra is the level induced by restricting the parent invariant form. We denote it by k_σ ; it equals the parent symbol k only after a trace-form convention has been fixed. In the standard non-simply-laced convention one must also record the long-root/short-root normalisation.

I.O.O.1. Orbifold route. Vol II catalogues the orbifold route: $\mathbb{Z}/n$ -invariants of an E_n -algebra (factorisation algebra, chiral algebra, vertex algebra) preserve the E_n -structure; the invariant subalgebra is an E_n -algebra whose spectral data descends from the parent. Coalgebra operations require the landing condition in (C4).

I.O.O.2. The four checks. For an order-two descended chiral coproduct to close inside $V_{k_\sigma}(\mathfrak{g}) \otimes V_{k_\sigma}(\mathfrak{g})$, four conditions must hold:

- (C1) **Potential σ -invariance.** The preprojective potential W on the parent σ -equivariant quiver is σ -invariant: $\sigma^*W = W$.
- (C2) **Hecke correspondence equivariance.** The Hecke correspondence on $\text{Rep}(\text{Dyn}(\mathfrak{g}^{\text{simply}}), \mathbf{d})$ is σ -equivariant: extensions of representations commute with Dynkin-diagram automorphisms.

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- (C₃) **Orbifold E_n -preservation.** Taking σ -invariants of an E_1 -chiral algebra produces an E_1 -chiral algebra. For a chiral coalgebra, equivariance produces a map to $(V \otimes V)^\sigma$; closure in $V^\sigma \otimes V^\sigma$ is a separate assertion. Once landing is known, the parent coassociativity and vertex-bialgebra/coaction identities restrict; without landing there is no coalgebra structure on the invariant object.
- (C₄) **Sign-sector vanishing.** The descended coproduct lands in $V_{k_\sigma}(\mathfrak{g}) \otimes V_{k_\sigma}(\mathfrak{g}) = V^+ \otimes V^+ \subset (V^+ \oplus V^-) \otimes (V^+ \oplus V^-)$; the sign-eigenspace-to-sign-eigenspace component $V^- \otimes V^-$ vanishes.

For the order-three triality fold the landing condition uses the character decomposition $V = \bigoplus_{\chi \in \widehat{\mathbb{Z}/3}} V^\chi$; diagonal invariance permits the neutral summands $V^{\chi_1} \otimes V^{\chi_2}$ and $V^{\chi_2} \otimes V^{\chi_1}$, and those summands must be killed by a separate landing calculation.

Definition 1.1 (Landing obstruction). Let Γ be a finite cyclic group acting by chiral bialgebra automorphisms on a chiral coalgebra (V, Δ) . Over characteristic zero write

$$V = \bigoplus_{\chi \in \widehat{\Gamma}} V^\chi.$$

Equivariance gives

$$\Delta(V^1) \subset (V \otimes V)^\Gamma = \bigoplus_{\chi\psi=1} V^\chi \otimes V^\psi.$$

The landing obstruction is

$$\mathcal{L}_\Gamma(\Delta) := \text{pr}_{\text{nt}} \circ \Delta|_{V^1} \in \text{Hom}\left(V^1, \bigoplus_{\chi \neq 1} V^\chi \otimes V^{\chi^{-1}}\right),$$

where pr_{nt} projects away from $V^1 \otimes V^1$. For completed chiral tensor products the same definition is made in the completed Hom-space.

Proposition 1.2 (Landing-obstruction descent criterion). *Let (V, Δ) be an E_1 -chiral bialgebra with an action of a finite cyclic group Γ by chiral bialgebra automorphisms. Then V^Γ inherits the E_1 -chiral algebra structure. The coproduct Δ restricts to a chiral coproduct*

$$V^\Gamma \longrightarrow V^\Gamma \otimes V^\Gamma$$

if and only if $\mathcal{L}_\Gamma(\Delta) = 0$. In that case the restricted coproduct is coassociative and satisfies the inherited vertex-bialgebra/coaction identities.

Proof. Finite invariants preserve algebra objects, so $V^\Gamma = V^1$ inherits the E_1 -chiral multiplication. Since Δ is Γ -equivariant, $\Delta(V^1)$ lies in $\bigoplus_{\chi\psi=1} V^\chi \otimes V^\psi$. It lies in $V^1 \otimes V^1$ exactly when the projection to the direct sum of the nontrivial neutral character sectors vanishes, which is precisely $\mathcal{L}_\Gamma(\Delta) = 0$. Coassociativity and the vertex-bialgebra identities are equations in the parent object; after landing, both sides of each equation lie in the fixed tensor powers and the equations restrict. $\square$

Remark 1.3 (Drinfeld–Sokolov compatibility). Folding is orthogonal to Drinfeld–Sokolov. DS reduction descends along a $\mathbb{C}$ -linear BRST differential from $V_k(\mathfrak{g})$ to $\mathcal{W}_k(\mathfrak{g}, f)$: it stays at fixed diagram and trades the loop direction for a conformal-weight grading. Folding, by contrast, descends along a finite-group action and produces a smaller Lie algebra at the same place in the Koszul hierarchy. The expected compatibility is a comparison between DS reduction of the fixed-current subalgebra and the corresponding invariant sector of $\mathcal{W}_k(\mathfrak{g}^{\text{simply}}, f_{\text{prin}})$. It is not used as a fixed-point theorem for the full orbifold vertex algebra. We use this compatibility only to compare folded chiral algebras with their Drinfeld–Sokolov reductions.

2. YANGIAN-LEVEL FOLDING

The Yangian-level folding fixes notation and gives the comparison object for any chiral lift.

Proposition 2.1 (Coproduct equivariance for the A_3 orbit fold). *Let σ be the A_3 Dynkin involution $\sigma(1) = 3, \sigma(2) = 2$. Then σ lifts to the RTT generators of $Y_h(\mathfrak{sl}_4)$ via $\sigma(T_{ij}(u)) = T_{\sigma(i), \sigma(j)}(u)$, the matrix coproduct $\Delta(T_{ij}(u)) = \sum_a T_{ia}(u) \otimes T_{aj}(u)$ is σ -equivariant. Hence*

$$\Delta(Y_h(\mathfrak{sl}_4)^\sigma) \subset (Y_h(\mathfrak{sl}_4) \otimes Y_h(\mathfrak{sl}_4))^\sigma.$$

Let $Y_h(\mathfrak{sl}_4) = Y^+ \oplus Y^-$ be the induced eigenspace decomposition. The obstruction to a fixed-subalgebra coproduct is

$$\mathcal{L}_\sigma^Y(\Delta) := \text{pr}_{Y^+ \otimes Y^-} \circ \Delta|_{Y^+} \in \text{Hom}(Y^+, Y^+ \otimes Y^-).$$

The fixed subspace carries the restricted matrix coproduct if and only if $\mathcal{L}_\sigma^Y(\Delta) = 0$. This is not the fixed-point assertion $Y_h(\mathfrak{sl}_4)^\sigma = Y_h(\mathfrak{sp}_4)$. The four-dimensional symplectic RTT presentation of $Y_h(\mathfrak{sp}_4)$ is a comparison target only after the folded/orbit relations, the spectral parameter, and the invariant-form normalisation have been fixed. Under the exceptional Lie algebra isomorphism $\mathfrak{sp}_4 \simeq \mathfrak{so}_5$, comparison with the five-dimensional orthogonal vector RTT realisation requires a separate normalisation of the P and Q terms.

Proof. The matrix coproduct equivariance is an index substitution:

$$\Delta(\sigma T_{ij})(u) = \Delta(T_{\sigma(i), \sigma(j)})(u) = \sum_a T_{\sigma(i), a}(u) \otimes T_{a, \sigma(j)}(u) = \sum_b \sigma(T_{ib}(u)) \otimes \sigma(T_{bj}(u))$$

after substituting $a = 5 - b$, and the right-hand side equals $(\sigma \otimes \sigma)\Delta(T_{ij})(u)$. This proves equivariance and hence the displayed inclusion into diagonal invariants. For an order-two action the diagonal invariant tensor square decomposes as $(Y^+ \otimes Y^+) \oplus (Y^- \otimes Y^-)$, so the projection to $Y^- \otimes Y^-$ is the exact obstruction to landing in $Y^+ \otimes Y^+$. Molev's type- C presentation supplies the symplectic RTT comparison only after that presentation and its normalisation data have been chosen. The exceptional identification with B_2 is an isomorphism of Lie algebras. The four-dimensional symplectic and five-dimensional orthogonal RTT modules remain different realisations. $\square$

Remark 2.2 (Orthogonal/symplectic RTT scope). Molev, Arnaudon–Molev–Ragoucy, and Guay–Regelskis–Wendlandt supply orthogonal and symplectic RTT presentations and coproducts [2–4]. They are comparison targets for the standard type- B and type- C folded constructions, not fixed-point theorems for ADE Yangian invariants. The RTT normalisation is part of the data: type B uses the orthogonal vector realisation and type C uses the symplectic defining realisation. These sources do not by themselves supply exceptional F_4 or G_2 RTT comparisons. Any such comparison must be imported from a separate exceptional Yangian source.

3. CHIRAL-LEVEL FOLDING: THE FOUR CHECKS

The chiral lift starts from the Jindal–Kaubrys–Latyntsev [1] critical CoHA vertex coproduct on the corresponding ADE quiver with potential. Identifying this CoHA object with the affine chiral algebra under discussion is an additional input. In the A_3 model the coproduct has the form

$$\Delta_{A_3}^v : V_k(\mathfrak{sl}_4) \longrightarrow V_k(\mathfrak{sl}_4) \otimes V_k(\mathfrak{sl}_4)$$

and one asks whether it descends to the invariant object $V_k(\mathfrak{sl}_4)^\sigma$. The first two checks are geometric; the sector-landing check, the induced level k_σ , and the spectral-parameter convention are independent.

(C1) σ -invariance of the preprojective potential.

Lemma 3.1 (σ -invariant preprojective potential). *The preprojective potential $W = \text{Tr}(aba'b' - a'b'ab)$ on the A_3 preprojective quiver is invariant under the Dynkin involution $\sigma(1) = 3, \sigma(2) = 2$, which lifts to the arrow action $\sigma: a \leftrightarrow b', \sigma: a' \leftrightarrow b$. By trace cyclicity, $\sigma^*W = W$.*

Proof. $\sigma^*W = \text{Tr}(\sigma(a)\sigma(b)\sigma(a')\sigma(b')) - \sigma(a')\sigma(b')\sigma(a)\sigma(b) = \text{Tr}(b'aba' - ba'b'a) = \text{Tr}(aba'b' - a'b'ab) = W$ by cyclicity. $\square$

(C2) Hecke correspondence equivariance.

Lemma 3.2 (Hecke correspondence is σ -equivariant). *The Hecke correspondence*

$$\text{Rep}(A_3, \mathbf{d}) \xleftarrow{p} \mathcal{Z} \xrightarrow{q} \text{Rep}(A_3, \mathbf{d}') \times \text{Rep}(A_3, \mathbf{d} - \mathbf{d}')$$

is σ -equivariant: the subrepresentation parameter $\mathcal{Z} = \{(M' \hookrightarrow M)\}$ carries the diagonal σ -action, and both projections commute with σ^ . The pushforward-pullback $q_! p^*$ commutes with σ^* at the level of constructible sheaves and at the level of vanishing cycles ϕ_W .*

Proof. Geometric naturality: if $(M' \hookrightarrow M)$ is a subrepresentation, so is $(\sigma M' \hookrightarrow \sigma M)$, because the Dynkin automorphism is a quiver automorphism. The vanishing cycle functor ϕ_W commutes with σ^* because $\sigma^*W = W$ by Lemma 3.1 and ϕ is natural in the pair (space, function). $\square$

(C3) Orbifold E_n -preservation.

Lemma 3.3 (Orbifold E_1 -chiral preservation). *If V is an E_1 -chiral algebra on a smooth curve X and $\sigma: V \rightarrow V$ is a finite-order automorphism, then the σ -invariants V^σ inherit an E_1 -chiral algebra structure. If V is also a chiral coalgebra with σ -equivariant coproduct Δ , then $\Delta(V^\sigma) \subset (V \otimes V)^\sigma$; it lands in $V^\sigma \otimes V^\sigma$ precisely when the nontrivial tensor-sector components vanish.*

Proof. The finite-orbifold lemma gives the algebra statement. Explicitly, the restriction of the chiral multiplication $\mu_z: V \boxtimes V \rightarrow \Delta_! V$ to $V^\sigma \boxtimes V^\sigma$ lands in $\Delta_!(V^\sigma)$ because the multiplication is σ -equivariant by construction. For the coalgebra statement, Proposition 1.2 gives the exact landing criterion. $\square$

(C4) Sign-sector vanishing.

Lemma 3.4 (Sign-sector vanishing criterion). *Let $V = V^+ \oplus V^-$ be the σ -eigenspace decomposition. The restriction $\Delta^v|_{V^+}$ descends to a coproduct on V^+ if and only if*

$$\mathcal{L}_{\mathbb{Z}/2}(\Delta^v) = \text{pr}_{V^- \otimes V^-} \circ \Delta^v|_{V^+} = 0.$$

Proof. For an order-two action, $(V \otimes V)^\sigma = (V^+ \otimes V^+) \oplus (V^- \otimes V^-)$. Thus a σ -equivariant coproduct on V restricts to a coproduct $V^+ \rightarrow V^+ \otimes V^+$ exactly when its projection to the second summand vanishes. The JKL vertex bialgebra identity and the VOA automorphism property preserve the eigenspace decomposition; they do not alone force the $V^- \otimes V^-$ projection to vanish. $\square$

Consequence: the folded chiral coproduct.

Proposition 3.5 (Landing-obstruction coproduct for C_2). *Assume the JKL critical CoHA vertex coproduct identifies with the chiral object attached to the A_3 affine input. Write $V_k(\mathfrak{sl}_4)^{\sigma, \text{cur}}$ for the affine subalgebra generated by the fixed current Lie algebra $(\mathfrak{sl}_4)^\sigma \simeq \mathfrak{sp}_4$; this is the image of $V_{k_\sigma}(\mathfrak{sp}_4)$ in the chosen affine quotient convention. The obstruction to a descended coproduct is*

$$\mathcal{L}_{\mathbb{Z}/2}(\Delta_{A_3}^v) \in \text{Hom}(V^+, V^- \otimes V^-).$$

The fixed-current image carries the descended coproduct if and only if $\mathcal{L}_{\mathbb{Z}/2}(\Delta_{A_3}^v) = 0$. When it vanishes, the map

$$\Delta_{C_2}^v := \Delta_{A_3}^v|_{V_k(\mathfrak{sl}_4)^{\sigma, \text{cur}}} : V_{k_\sigma}(\mathfrak{sp}_4) \longrightarrow V_{k_\sigma}(\mathfrak{sp}_4) \otimes V_{k_\sigma}(\mathfrak{sp}_4)$$

is a chiral coproduct on $V_{k_\sigma}(\mathfrak{sp}_4)$, coassociative and compatible with the vertex operator map. If, in addition, the matrix-element evaluation is taken in the same spectral parameter and invariant-form convention as the symplectic RTT presentation, it compares with the RTT matrix coproduct of Proposition 2.1.

Proof. Lemmas 3.1–3.4 and Proposition 1.2 give the descent criterion: (C1) and (C2) combine to give σ -equivariance of $\Delta_{A_3}^v$ at the level of the vanishing-cycle sheaf; (C3) ensures the fixed-current subalgebra inherits the chiral algebra structure; the vanishing of $\mathcal{L}_{\mathbb{Z}/2}(\Delta_{A_3}^v)$ is exactly the statement that the coproduct closes inside $V^+ \otimes V^+$. The parent coassociativity and vertex-bialgebra identities then restrict. The RTT matrix-element comparison is exactly the remaining normalisation input: the spectral variable, the $\hbar$ convention, and the induced non-simply-laced invariant form must agree with the chosen symplectic RTT presentation. $\square$

4. EXTENSION: B_N, C_N, F_4

The same four checks apply to the remaining non-triviality foldings.

Theorem 4.1 (Landing-obstruction theorem for B_N, C_N, F_4). *Let $(\mathfrak{g}^{\text{simply}}, \sigma, \mathfrak{g})$ run over the pairs $(D_{N+1}, \mathbb{Z}/2, B_N)$, $(A_{2N-1}, \mathbb{Z}/2, C_N)$, and $(E_6, \mathbb{Z}/2, F_4)$, and let k_σ be the induced level on the fixed current algebra. The three geometric checks construct a σ -equivariant parent chiral coproduct, and the obstruction to a folded coproduct is the Hom-valued class*

$$\mathcal{L}_{\mathbb{Z}/2}(\Delta_{\mathfrak{g}^{\text{simply}}}^v) \in \text{Hom}(V^+, V^- \otimes V^-).$$

More explicitly:

- (a) The preprojective potential on the corresponding double quiver is σ -invariant (trace cyclicity);
- (b) The Hecke correspondence is σ -equivariant (geometric naturality);
- (c) Orbifold E_1 -chiral preservation applies (the orbifold-route lemma);
- (d) The sector-vanishing condition of Lemma 3.4 is an additional explicit calculation, equivalently $\mathcal{L}_{\mathbb{Z}/2}(\Delta_{\mathfrak{g}^{\text{simply}}}^v) = 0$.

If the JKL-to-affine identification holds, the fixed-current image of $V_{k_\sigma}(\mathfrak{g})$ carries a chiral coproduct $\Delta_{\mathfrak{g}}^v$ if and only if $\mathcal{L}_{\mathbb{Z}/2}(\Delta_{\mathfrak{g}^{\text{simply}}}^v) = 0$. This does not identify the full invariant vertex algebra with the fixed-current affine algebra unless a separate generation theorem is supplied. For the B and C series, comparison with the orthogonal or symplectic RTT matrix coproduct is a separate Molev-type input and requires the spectral-parameter and root-length normalisations. No F_4 RTT comparison is asserted without an exceptional source.

Proof. Each of (a)–(c) is the relevant extension of the A_3 argument of §3. Trace cyclicity in (a) is diagram-independent; Hecke equivariance in (b) is geometric; orbifold preservation in (c) is via the orbifold-route lemma. Proposition 1.2 identifies the remaining obstruction group and proves the if-and-only-if descent statement. When the obstruction vanishes, coassociativity and the coaction identities restrict from the parent coproduct. $\square$

Remark 4.2 (Matching with Drinfeld–Sokolov). There is a natural comparison between DS reduction of the fixed-current subalgebra and the invariant sector of the principal $\mathcal{W}$ -algebra $\mathcal{W}_k(\mathfrak{g}^{\text{simply}}, f_{\text{prin}})$ when σ preserves the principal nilpotent. If this BRST/invariant comparison is a quasi-isomorphism

in the chosen level convention, the coproduct constructed by Theorem 4.1 descends to the folded $\mathcal{W}$ -algebra. This is an additional compatibility input, not an automatic fixed-point theorem for the full orbifold $\mathcal{W}$ -algebra.

5. G_2 : THE TRIALITY OBSTRUCTION

The folding $G_2 = (D_4)^{\mathbb{Z}/3}$ is the order-3 Dynkin automorphism permuting the three outer legs. Two genuine obstructions appear.

Proposition 5.1 (Triality landing obstruction for G_2). *Let $u \in V^{\chi_1}$ and $v \in V^{\chi_2}$ be the two spin-four triality fields in the D_4 parent, and suppose that the leading coproduct on this doublet is primitive:*

$$\Delta(u) = u \otimes 1 + 1 \otimes u, \quad \Delta(v) = v \otimes 1 + 1 \otimes v$$

modulo lower-spin fixed fields. Then the invariant polynomial sector

$$\mathbb{C}[u, v]^{\mathbb{Z}/3} = \mathbb{C}[x, y, z]/(xy - z^3), \quad x = u^3, \quad y = v^3, \quad z = uv,$$

does not inherit the restricted parent coproduct. The obstruction

$$\mathcal{L}_{\mathbb{Z}/3}(\Delta_{D_4}^v) \in \text{Hom}(V^{\chi_0}, V^{\chi_1} \otimes V^{\chi_2} \oplus V^{\chi_2} \otimes V^{\chi_1})$$

is nonzero unless an additional quotient, coideal, or modified folded coproduct kills the displayed nontrivial neutral character sectors.

Proof. The diagonal $\mathbb{Z}/3$ action is the symplectic action of type $\frac{1}{3}(1, 2)$ on the (u, v) -plane. Hence the invariant ring is the A_2 du Val surface $\mathbb{C}[u^3, v^3, uv] \cong \mathbb{C}[x, y, z]/(xy - z^3)$. The primitive coproduct gives

$$\begin{aligned} \Delta(z) &= z \otimes 1 + u \otimes v + v \otimes u + 1 \otimes z, \\ \Delta(x) &= x \otimes 1 + 3u^2 \otimes u + 3u \otimes u^2 + 1 \otimes x, \\ \Delta(y) &= y \otimes 1 + 3v^2 \otimes v + 3v \otimes v^2 + 1 \otimes y. \end{aligned}$$

The middle terms of $\Delta(z)$ lie in $V^{\chi_1} \otimes V^{\chi_2} \oplus V^{\chi_2} \otimes V^{\chi_1}$. The middle terms of $\Delta(x)$ and $\Delta(y)$ lie in the same nontrivial neutral summands after reducing characters modulo 3. These tensor factors are diagonal invariants but not elements of $V^{\chi_0} \otimes V^{\chi_0}$. Therefore the parent coproduct lands in $(V \otimes V)^{\mathbb{Z}/3}$ and fails to land in the tensor square of the fixed sector. This is exactly $\mathcal{L}_{\mathbb{Z}/3}(\Delta_{D_4}^v) \neq 0$ on the primitive triality doublet. $\square$

5.0.0.1. Obstructions.

(O1) *Cyclic quotient singularity.* The natural triality doublet has local action of type $\frac{1}{3}(1, 2)$, not $\frac{1}{3}(1, 1)$. Thus $\mathbb{C}^2/\mathbb{Z}_3$ is the A_2 du Val surface and admits a crepant resolution. The singular geometry is not the obstruction to the coproduct. Proposition 5.1 identifies the obstruction as landing: the invariant ring is crepant, but the primitive parent coproduct still produces nontrivial neutral character sectors.

(O2) *Third roots of unity in the spectral parameter.* The JKL vertex bialgebra axiom $\Delta^v(Y(a, z)b) = Y^{(2)}(\Delta^v(a), z)\Delta^v(b)$, twisted by $\mathbb{Z}/3$, requires a $\mathbb{Z}/3$ -equivariant enhancement with a spectral twist by cube roots of unity. The sign-sector decomposition $V = V^+ \oplus V^-$ of the order-2 case generalises to a character decomposition $V = V^{\chi_0} \oplus V^{\chi_1} \oplus V^{\chi_2}$; the vanishing lemma (Lemma 3.4) now requires vanishing of every component except $V^{\chi_0} \otimes V^{\chi_0}$. In particular the neutral nontrivial summands $V^{\chi_1} \otimes V^{\chi_2}$ and $V^{\chi_2} \otimes V^{\chi_1}$ are diagonal $\mathbb{Z}/3$ -invariants but must still vanish for the coproduct to land in $V^{\chi_0} \otimes V^{\chi_0}$.

The Yangian-level triality comparison is a folded/orbit presentation, not a general fixed-point equality. The chiral lift requires the triality-sector calculation and a genuine G_2 Yangian source for the RTT comparison.

6. SUMMARY

- (1) **Yangian level.** The $\mathcal{A}_3$ coproduct equivariance is explicit (Proposition 2.1); comparison with type C_2 uses a folded/orbit presentation plus spectral and invariant-form normalisation. The orthogonal and symplectic RTT sources cover the B and C comparison targets, while F_4 and G_2 require separate exceptional sources.
- (2) **Order-2 chiral descent.** Theorem 4.1: the obstruction to a folded chiral coproduct for B_N, C_N, F_4 is the class $\mathcal{L}_{\mathbb{Z}/2}(\Delta_{\mathfrak{g}^{\text{simply}}}^v)$ in $\text{Hom}(V^+, V^- \otimes V^-)$; its vanishing is necessary and sufficient after the JKL-to-affine identification, with induced level and root-length normalisation fixed.
- (3) **Triality case.** $G_2 = (D_4)^{\mathbb{Z}/3}$ (Proposition 5.1); the primitive triality doublet has a nonzero landing obstruction. A folded G_2 coproduct therefore requires an additional quotient, coideal, or modified coproduct plus the cube-root spectral convention.

The four checks are the folding discipline: potential invariance, Hecke equivariance, finite-orbifold E_1 preservation, and sector vanishing. The first three are geometric. The fourth is the deciding coproduct calculation.

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